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Gauss-Seidel

The Gauss-Seidel method is an iterative technique for solving a system of linear equations. The method improves upon the Jacobi method by using the most recent updates of the variables as soon as they are available.

Steps of the Gauss-Seidel Method:

- Formulate the System of Equations: Start with a system of linear equations in the form $Ax = b$:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases}$$

- Rewrite Each Equation: Rewrite each equation to solve for one of the variables in terms of the others. For example, solve the i-th equation for x_i:

$$x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j \right)$$

- Initial Guess: Choose an initial guess for the values of the unknowns $(x_1^{(0)}, x_2^{(0)}, \ldots, x_n^{(0)})$.

- Iterative Process: Update each variable sequentially using the most recent values. For the k-th iteration:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right)$$

  Here, $x_j^{(k+1)}$ is the updated value and $x_j^{(k)}$ is the previous iteration's value.

- Convergence Check: Continue the iterations until the solution converges, i.e., the changes in the variables between successive iterations are smaller than a chosen tolerance level:

$$|x_i^{(k+1)} - x_i^{(k)}| < \epsilon \quad \text{for all } i$$

Problem

Given the system of equations, find the solution approximation after 1 iteration of the Gauss-Seidel method:

$$\begin{cases} 60x_1 - 30x_2 - 20x_3 = -400 \\ -30x_1 + 180x_2 - 60x_3 = 400 \\ -20x_1 - 60x_2 + 150x_3 = 300 \end{cases}$$

we can rewrite the equations in the form suitable for the method. Starting with an initial approximation $(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}) = (0, 0, 0)$:

1. **Initial Values:**

$$(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}) = (0, 0, 0)$$

2. **Step 1:** Write the equations in a form suitable for the Gauss-Seidel method

   The given system of equations, we need to rewrite each equation to solve for one of the variables in terms of the others:

$$x_1 = \frac{1}{60}(-400 + 30x_2 + 20x_3)$$

$$x_2 = \frac{1}{180}(400 + 30x_1 + 60x_3)$$

$$x_3 = \frac{1}{150}(300 + 20x_1 + 60x_2)$$

3. **Step 2**: Iterative Process

   We use the Gauss-Seidel update rules to compute the next approximations. We will perform one iteration to demonstrate the process.

   - **Iteration 1**:

     1. Update x_1:

$$x_1^{(1)} = \frac{-400 + 30x_2^{(0)} + 20x_3^{(0)}}{60} = \frac{-400 + 30 \cdot 0 + 20 \cdot 0}{60} = \frac{-400}{60} = -6.6667$$

     2. Update x_2:

$$x_2^{(1)} = \frac{400 + 30x_1^{(1)} + 60x_3^{(0)}}{180} = \frac{400 + 30 \cdot -6.6667 + 60 \cdot 0}{180} = \frac{400 - 200}{180} = 1.1111$$

3. Update x_3:

$$x_3^{(1)} = \frac{300 + 20x_1^{(1)} + 60x_2^{(1)}}{150} = \frac{300 + 20 \cdot -6.6667 + 60 \cdot 1.1111}{150}$$

$$x_3^{(1)} = \frac{300 - 133.334 + 66.666}{150} = 1.1111$$

- **Iteration 2**

1. Update x_1:

$$x_1^{(2)} = \frac{-400 + 30x_2^{(1)} + 20x_3^{(1)}}{60} = \frac{-400 + 30 \cdot 1.1111 + 20 \cdot 1.1111}{60}$$

$$x_1^{(2)} = \frac{-400 + 33.333 + 22.222}{60} = -5.7245$$

2. Update x_2:

$$x_2^{(2)} = \frac{400 + 30x_1^{(2)} + 60x_3^{(1)}}{180} = \frac{400 + 30 \cdot -5.7245 + 60 \cdot 1.1111}{180}$$

$$x_2^{(2)} = \frac{400 - 171.735 + 66.666}{180} = 1.6213$$

3. Update x_3:

$$x_3^{(2)} = \frac{300 + 20x_1^{(2)} + 60x_2^{(2)}}{150} = \frac{300 + 20 \cdot -5.7245 + 60 \cdot 1.6213}{150}$$

$$x_3^{(2)} = \frac{300 - 114.49 + 97.278}{150} = 1.606$$

- **Iteration 3** 1. Update x_1:

$$x_1^{(3)} = \frac{-400 + 30x_2^{(2)} + 20x_3^{(2)}}{60} = \frac{-400 + 30 \cdot 1.6213 + 20 \cdot 1.606}{60}$$

$$x_1^{(3)} = \frac{-400 + 48.639 + 32.12}{60} = -5.3184$$

2. Update x_2:

$$x_2^{(3)} = \frac{400 + 30x_1^{(3)} + 60x_3^{(2)}}{180} = \frac{400 + 30 \cdot -5.3184 + 60 \cdot 1.606}{180}$$

$$x_2^{(3)} = \frac{400 - 159.552 + 96.36}{180} = 1.8755$$

3. Update x_3:

$$x_3^{(3)} = \frac{300 + 20x_1^{(3)} + 60x_2^{(3)}}{150} = \frac{300 + 20 \cdot -5.3184 + 60 \cdot 1.8755}{150}$$

$$x_3^{(3)} = \frac{300 - 106.368 + 112.53}{150} = 2.0558$$

- Results after 3 Iterations:

$$(x_1^{(3)}, x_2^{(3)}, x_3^{(3)}) = (-5.3184, 1.8755, 2.0558)$$

These values are the approximations of x_1, x_2, and x_3 after 3 iterations of the Gauss-Seidel method starting from the initial guess $(0, 0, 0)$. This method converges towards the solution, and the accuracy improves with more iterations.